

GPA Calculator Project Report

This project focuses on building a simple GPA calculator using Python. The idea came from the difficulty many students face when trying to understand how their grades are calculated, especially in systems where relative grading is used instead of fixed percentage-based evaluation.

Unlike basic calculators, this program does not just take marks and convert them directly into grades. Instead, it compares a student's performance with the class average and the highest score in the class. This makes the result more realistic and closer to how grading is actually done in many universities.

The program runs in a command-line interface and interacts with the user step by step. It first asks for general details such as the student's name and year of study. Then it collects information about each subject, including marks scored, class average, and highest marks. Based on these inputs, the program assigns grades like S, A, B, C, D, E, or F.

One interesting part of the project is how the grading logic works. Instead of using fixed cutoffs, it checks how close a student is to the top score or how far they are from the average. This approach makes the grading system more dynamic and fair in competitive environments.

The project is written entirely in Python, which makes it easy to understand and run. No external libraries are required, and the code uses basic concepts such as loops, conditional statements, and lists. This makes it a good project for beginners who want to apply programming concepts to real-world problems.

During testing, different sets of values were entered to see how the system behaves. For example, when the student's marks were very close to the highest marks, the program correctly assigned an S grade. Similarly, lower scores resulted in appropriate lower grades. This helped confirm that the logic is working as expected.

Overall, this project helped in understanding both programming and academic grading systems better. It shows how even a small piece of software can make everyday tasks easier and more accurate. With further improvements, this tool could be expanded into a full application with a graphical interface or additional features like CGPA tracking.

In conclusion, the GPA calculator is a useful and practical project that combines logical thinking with real-life application. It not only simplifies grade calculation but also gives students a clearer idea of where they stand academically.